Why (and How) LGADs Work: Ionization, Space Charge, and Gain Saturation

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Abstract

The temporal resolution of Low-Gain Avalanche Detectors (LGADs), also known as Ultra-Fast Silicon Detectors (UFSDs), is governed by two contributions: jitter, arising from electronic noise and signal slew rate, and the Landau noise term, arising from the non-uniform energy deposition of minimum ionizing particles (MIPs). We show that a correct simulation of the initial ionization alone significantly overestimates the measured Landau noise. Two additional physical mechanisms are necessary to reproduce the data: space charge effects during electron/hole drift, which smooth the granularity of the initial charge distribution, and gain saturation during multiplication, which preferentially suppresses large-amplitude fluctuations. All steps of the model have been implemented in the fast simulation program Weightfield2 (WF2). The model is validated against several independent experimental observations: the evolution of the measured charge distribution with gain, the temporal resolution of events in the Landau tail, and the thickness dependence of timing performance. We also discuss a data-driven gain measurement method based on gain saturation, and implications for gain layer design.

Keywords: LGAD; UFSD; temporal resolution; Landau noise; gain saturation; space charge; Weightfield2; silicon detectors

1. Introduction

Low-Gain Avalanche Detectors (LGADs) [1, 2] are silicon sensors that exploit a moderate internal gain (~10–40) to achieve excellent single-layer temporal resolution, routinely below 50 ps [4]. Their timing performance is central to several detector upgrades at the High-Luminosity LHC, including the CMS Endcap Timing Layer (ETL) and the ATLAS High-Granularity Timing Detector (HGTD) [5, 6].

Despite their widespread use, a complete quantitative understanding of *why* LGADs achieve such good timing is still being developed. This work addresses that question from first principles, tracing the signal formation chain from the initial ionization through drift and multiplication, and identifying the physical mechanisms that govern the intrinsic temporal resolution.

All simulation results presented in this paper were obtained with the Weightfield2 (WF2) program [7], a fast, publicly available simulation tool for silicon sensors available at `https://www.to.infn.it/~cartigli/Weightfield2/`.

The paper is organized as follows. Section 2 describes the MIP ionization model, including the seed distribution and long-range correlations from delta rays. Section 3 introduces the two components of temporal resolution and provides an analytical derivation of the Landau noise term. Section 4 presents the comparison between WF2 predictions and data, identifying the discrepancy that motivates two additional physical mechanisms. Section 5 describes the two smoothing mechanisms — space charge effects during drift and gain saturation during multiplication — and introduces the phenomenological parameter α. Section 6 presents experimental evidence used to determine α. Section 7 shows that, with α fixed, WF2 predictions agree with measured temporal resolution across all sensor thicknesses. Section 8 summarises why LGADs achieve good timing through the interplay of these three mechanisms. Section 9 introduces a novel data-driven gain measurement method. Section 10 discusses implications for

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gin layer design. Section 11 summarizes the conclusions.

2. MIP Ionization in Silicon Sensors

2.1. The Seed Distribution Model

The energy deposition of a MIP in a silicon sensor of arbitrary thickness is modeled in WF2 as a sum of local deposits drawn independently from a *seed distribution* — the energy deposited in a 1-μm thick silicon layer. This seed distribution has two physically distinct components:

- A soft scattering component, corresponding to small, frequent energy transfers. Because the cross-section diverges at low T, the vast majority of interactions transfer very little energy. In WF2, these are modeled as a Gaussian contribution with mean $\mu_{\text{soft}}(d)$ and standard deviation σ_{soft} per micron slice.
- A hard scattering component, corresponding to more energetic collisions sampled from the $1/T^2$ distribution over $[T_{\min}, T_{\max}]$, that produce δ-rays — knock-on electrons carrying significant kinetic energy away from the primary track. The value of $T_{\min}$ sets the width of this energy range and hence the normalisation of the distribution. A δ-ray with $T \sim 30$–100 keV can travel tens of microns from the primary track. These events populate the high-energy Landau tail.

Concretely, in each 1-μm slice the number of hard collisions N is drawn from a Poisson distribution with mean $\nu \cdot \Delta x$, where ν is the mean collision rate per micron. Each collision energy is sampled from the truncated $1/T^2$ distribution using the exact inverse-CDF method:

$$T = \frac{1}{1/T_{\min} - u\,(1/T_{\min} - 1/T_{\max})}, \qquad u \sim \mathcal{U}(0,1) \tag{1}$$

with $T_{\min} = 26$ eV and $T_{\max} = 600$ keV. The results are insensitive to $T_{\max}$ for values above ~ 200 keV, as the $1/T^2$ spectrum strongly suppresses large energy transfers. The soft component is added as an independent Gaussian draw (μ_{soft}, $\sigma_{\text{soft}} = 10$ eV) per micron. The total slice energy (hard + soft) is converted to electron-hole pairs using the mean ionisation energy in silicon, $\epsilon = 3.6$ eV/pair. Both ν and μ_{soft} depend on sensor thickness following empirical scaling relations:

$$\nu(d) = \frac{3.8}{d^{0.43}}, \qquad \mu_{\text{soft}}(d) = 22\ln(d) - 30\ [\text{eV}] \tag{2}$$

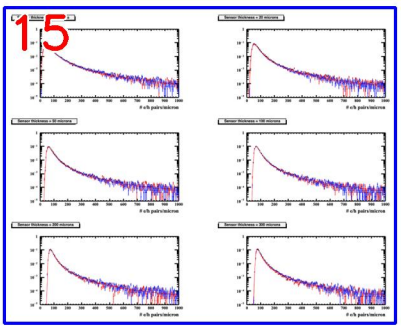

Figure 1: Measured (red) and WF2-simulated Landau energy distributions (blue). The simulation correctly reproduces both the MPV and the FWHM as a function of sensor thickness.

This thickness dependence reflects the fact that the relative weight of the hard and soft components in the energy deposition changes systematically with thickness. The scaling functions were determined by fitting the measured MPV and FWHM of the Landau distribution across sensor thicknesses from 5 to 300 μm. The total energy deposition in a sensor of thickness d is obtained by drawing a sequence of $d/1\,\mu$m random samples from the seed distribution and summing them. Using this approach, WF2 correctly reproduces the measured MPV [8] and FWHM of the Landau distribution as a function of sensor thickness (Figure 1), confirming the validity of the ionization model.

2.2. Long-Range Correlations from Delta Rays

In WF2, each δ-ray of energy T is assigned a range using the empirical CSDA relation for electrons in silicon [10]:

$$r[\mu\text{m}] = 0.04 \cdot \left(\frac{T}{1\ \text{keV}}\right)^{1.75} \tag{3}$$

δ-rays with $R < 2\,\mu$m deposit their energy locally at the point of creation. For $R \geq 2\,\mu$m, the energy is distributed along the track using a power-law stopping profile derived by inverting Eq. (3) ($R \propto T^{1.75}$ implies $T \propto R^{1/1.75}$):

$$T_{\text{remaining}}(x) = T \cdot \left(\frac{R - x}{R}\right)^{1/1.75} \tag{4}$$

so that the energy deposited in each 1-μm slice is the difference in remaining energy between its start and end. This reflects the physical behaviour of electrons, whose energy loss per unit length increases as they slow down, so that the deposited energy per slice rises toward the end of the range. If the range of a δ-ray exceeds the remaining distance to the sensor edge, energy deposition

is truncated at the boundary and the remaining energy escapes the sensor undetected.

3. Components of the Temporal Resolution

3.1. Jitter and Landau Noise

The temporal resolution σ_t of an LGAD is decomposed into two independent contributions [3]:

$$\sigma_t^2 = \sigma_{\text{jitter}}^2 + \sigma_{\text{Landau}}^2 \tag{6}$$

Jitter arises from electronic noise σ_V and the signal slew rate dV/dt at threshold crossing:

$$\sigma_{\text{jitter}} = \frac{\sigma_V}{dV/dt} \tag{9}$$

It dominates at low gain, where the signal amplitude is small relative to the noise.

Landau noise, also called the non-uniform ionization term, arises from the event-to-event variability of the charge deposition. Each MIP deposits charge in a different spatial pattern along the track, producing a different signal shape and hence a different trigger time. This term dominates at high gain, where jitter is negligible.

Throughout this work, the trigger time is defined by a Constant Fraction Discriminator (CFD) at 30% of the signal amplitude, both in the experimental data and in the WF2 simulation.

3.2. Analytical Derivation of the Landau Noise Term

Following Riegler [9], we derive the general features of the Landau noise term. Consider a sensor of thickness d in which a single electron-hole pair is created at a random, uniformly distributed position $x \in [0, d]$. The electron drifts to the collecting electrode with velocity v, arriving at time $t = x/v$. Since x is uniform on $[0, d]$, the standard deviation of the arrival time is:

$$\sigma_{\text{Landau}} = \frac{d}{v\sqrt{12}} \tag{16}$$

This is the fundamental upper bound on the Landau noise: it corresponds to total ignorance of the charge creation position. For a laser that deposits N electron-hole pairs uniformly along the track, the uncertainty on the trigger time improves as $1/\sqrt{N}$ (by analogy with the mean of N uniform random variables).

$$\sigma_{\text{Laser}} = \frac{\sqrt{d}}{v\sqrt{12n}} \tag{19}$$

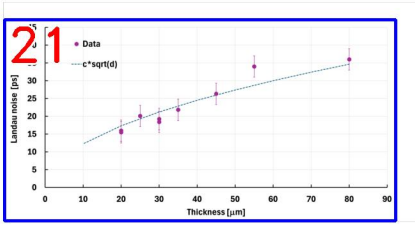

Figure 2: Measured values of the Landau noise as a function of sensor thickness and a line that shows the $\propto \sqrt{d}$ behaviour.

where n is the linear density of electron-hole pairs (pairs per unit length). For the MIP case, the charge deposition is non-uniform and the analytical treatment becomes significantly more complex; the full derivation is given in [9].

Equation (8) reveals the fundamental structure of the Landau noise: it grows with the square root of the thickness and decreases with drift velocity. Operating at fields sufficient to saturate the drift velocity v, and reducing sensor thickness, are the most effective ways to reduce this term. Specifically, for the MIP case, the N charge carriers are not uniformly distributed but follow the non-uniform Landau deposition. The resulting Landau noise is larger than for a laser, and its precise value depends on the specific ionization pattern in each event [9]. Figure 2 shows that the theoretical $\sqrt{d}$ scaling well matches the measured values of Landau noise [11].

4. The Discrepancy Between Simulation and Data

Figure 3 compares the measured temporal resolution (jitter subtracted, Landau noise contribution only) with the WF2 predictions obtained using the Landau ionization model described above. Two key observations emerge:

1. The simulated Landau noise is considerably larger than the measured one across all sensor thicknesses.
2. The discrepancy grows with increasing sensor thickness.

This discrepancy implies that the actual LGAD signal shape is smoother than the ionization model implemented in WF2. Two physical mechanisms, not included in the simulation, are responsible for this discrepancy: space charge effects during drift and gain saturation during multiplication.